Pokhara University
School of Engineering
Internal Examination

Level: Master, Semester: I
Course Title: Software Engineering
Program: Computer Science/ Computer Engineering

Full Marks: 100
Pass Mark: 60
Time: 4 hrs

Attempt all Questions

Q1.

(a) Reflect on the Vasa Tragedy and its implications for software engineering. How can changing requirements and poor communication lead to project failure? Which software development model (e.g., Waterfall, Agile, Spiral) would you recommend to handle such situations, and why? Discuss how modern project management practices could have mitigated the issues faced in the Vasa project. [8 Marks]

(b) What is the role of a DevOps culture and CI/CD (Continuous Integration/Continuous Deployment) in modern software development? How do they improve software delivery speed and reliability? [7 Marks]

Q2.

(a) Project organization involves defining roles, task distribution, and communication practices. Explain how these factors interact to influence project success. Give one example of a communication breakdown and how it could be avoided. [8 Marks]

OR

Consider a scenario where a software project has conflicting requirements from different stakeholders (e.g., end-users, developers, and project managers). How would you prioritize and manage these conflicts to ensure project success? What techniques or tools would you use to facilitate negotiation and consensus?

(b) Explain the concept of containerization. How does Docker simplify application deployment across different environments? Mention one benefit over traditional virtual machines. [7 Marks]

Q3. *(15 Marks)*

For an Online Bookstore Platform with customers, bookstore owners, and delivery personnel:

(a) List the key actors and stakeholders. For each, describe their main interests in the system. [5 Marks]

(b) Create a use case for "Purchase a Book" (customer) and a use case for "Add New Book to Catalog" (bookstore owner). Include the basic flow of events. [5 Marks]

(c) Draw a use case diagram showing the actors and the following use cases: Purchase Book, Manage Catalog, Track Delivery. Explain how the diagram helps capture stakeholder needs. [5 Marks]

Q4.

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A startup is developing a Smart Healthcare Management System that allows patients to book appointments, doctors to manage schedules, and hospitals to maintain medical records. The system must be scalable, easy to maintain, and adaptable to future features such as telemedicine and AI-based diagnosis. During the design phase, the software team decides to follow Object-Oriented Analysis and Design (OOAD) principles to ensure modularity, reusability, and flexibility. They need to model entities like Patient, Doctor, Appointment, and MedicalRecord while keeping the system easy to extend and maintain.

(a) Using the above scenario, define the four core principles of Object-Oriented Analysis and Design (OOAD): encapsulation, abstraction, inheritance, and polymorphism. [8 Marks]

(b) Consider a scenario where you are designing a real-time IoT system for a smart city. How would you leverage Edge Computing to address challenges like network latency and limited resources? Discuss the role of Edge AI in enabling real-time decision-making and optimizing data processing. [7 Marks]

Q5.

(a) What is the primary purpose of the Factory Method design pattern? How does it promote flexibility in software development, and what are the potential drawbacks of using this pattern?

Or

What is the Liskov Substitution Principle (LSP)? Explain with an example where violating LSP could cause issues in a software system, and how adhering to LSP can improve system design

(b) Define the term "Design Pattern" in the context of software engineering. How do design patterns help in software development, and what are the three main categories of design patterns? How do patterns promote software reuse and maintainability? [8 Marks]

Q6.

(a) A software company is developing an Online Food Delivery System that includes features such as user registration, restaurant listings, order placement, payment processing, and delivery tracking. During testing, the team discovers that while individual modules seem to work correctly, issues arise when modules interact, and some system-level failures occur under real-world conditions. The project manager wants to ensure a robust testing strategy that can identify defects early, validate interactions between components, and confirm that the complete system meets user expectations.

- Define unit testing, integration testing, and system testing.
- Analyze at which stage of the software development lifecycle each type of testing should be performed and justify why.
- Evaluate the primary purpose of each testing level in the context of the system described.
- Critically discuss how these three levels of testing work together to ensure overall software quality and reliability [8 Marks]

(b) Evaluate the use of version control systems (VCS) in managing requirements documents. How does maintaining a central repository with version control improve collaboration, reduce risks, and ensure compliance in large-scale projects? Discuss the role of audit trails, change tracking, and backup solutions in this context. [7 Marks]

Q7. Write short notes on any two of the following *(each carries 5 marks)*:

(a) Service-Oriented Architecture (SOA)

(b) Secure Coding Practices

(c) Software Quality Assurance (SQA) Activities

(d) Internet of Things (IoT) based Software Engineering